Online on xv6 System Call

Section: C2 — **Time:** 25 minutes

In this assignment, you will add system calls to store a single-byte shift key in kernel memory and apply a standard Caesar cipher transformation to a string payload provided by a user application via a struct pointer.

Formally add 2 system calls:

1. `setCipherKey(int key)` : Sets the internal kernel shift key (a value between 1 and 25).
2. `transformBuffer(struct msg_buffer *buf)` : Takes a pointer to a `msg_buffer` struct and shifts every alphabetic character in `buf->data` forward by `key` positions (wrapping 'z' to 'a' and 'Z' to 'A'). Non-alphabetic characters remain unchanged. Updates `buf->data` in place.

The structure `msg_buffer` is defined as follows:

```
struct msg_buffer {
    int len;       // length of string
    char data[32]; // string buffer
};
```

You also need to add 2 user commands:

3. `cipher_key k`
4. `cipher_run text`

See the sample I/O for clarification:

```
$ cipher_key 3
Cipher key set to 3
$ cipher_run hello
Transformed: khoor
$ cipher_key 1
Cipher key set to 1
$ cipher_run Zebra
Transformed: Afbsb
$ cipher_run xv6
Transformed: yw6
```

Submission:

```
git add --all
git diff HEAD > ../{studentID}.patch
```